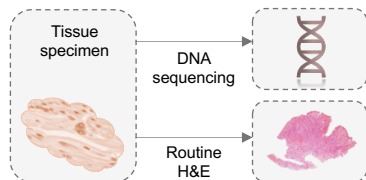
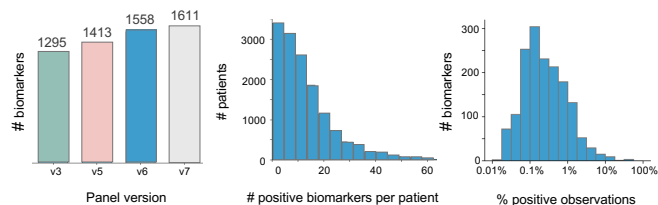


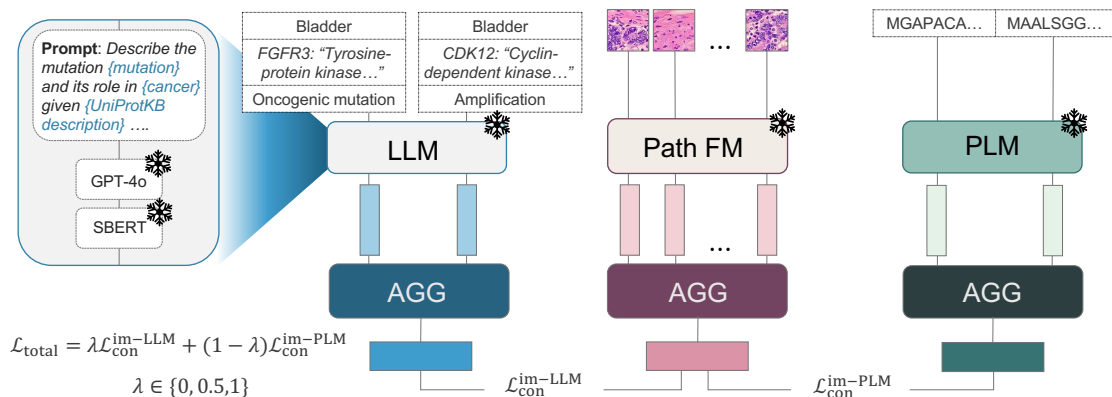
a. Typical workflow



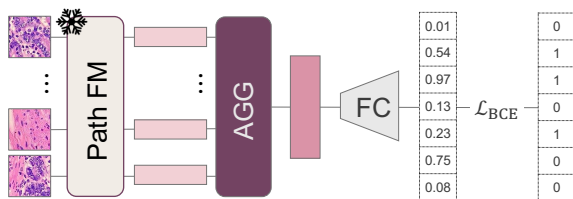
b. MSK-IMPACT panel statistics



c. MARBLE pretraining approach



d. Biomarker prediction: Supervised fine-tuning



e. Experimental cohorts

